
INFO2000

Assignment 5

Exploratory Data Analysis (EDA) using Pandas

Instructions

This is an assignment on Exploratory Data Analysis. It includes data cleaning and data analysis and will test your pandas and plotting skills using the different packages you have been exposed to. While the videos cover many topics in pandas and plotting, looking into the documentation and using all the possibilities to solve the problems will be important and it will serve you well in your informatics career.

Dataset

The dataset for this project is called 'covid_data.csv'. The dataset gives a daily array of records for 3 years and is split by country and continent. The meaning/type of information, for each variable, is provided as a separate document called 'Columns_Info.pdf'. You will drop many of these columns during the data cleaning process.

1 Pandas

1. Read in the data set into a pandas DataFrame.
2. Gather all the information about the dataset and its statistics using the appropriate Pandas commands. Check the shape of the dataset before and after preprocessing.
3. **Preprocessing the data:** Please do all the preprocessing in one cell and let the resulting dataframe be the master dataframe for the project.
 - (a) Remove all columns having 60% or more missing values.
 - (b) Remove all rows which have missing values.
 - (c) Drop all columns which have the word 'smoothed' in their column names.
 - (d) Convert date to datetime datatype and set it as index.
 - (e) Check and confirm that there are no more missing values in the resulting dataset
4. Plot the following high-level charts to get an idea of the various developments on a quarterly and monthly basis. Plot 2 charts in each row.
 - (a) Daily new cases per million on a worldwide basis
 - (b) Daily new deaths per million on a worldwide basis
 - (c) Total cases per million on a quarterly basis
 - (d) Total deaths per million on a quarterly basis
5. Plot the following:
 - (a) A single chart showing the country wise (use 'location' or 'iso_code') daily development of new cases per million for USA, India, Russia, France, and China.
6. Form a new dataframe by grouping the data by 'iso_code', which are basically country codes, and do the following:
 - (a) Plot the total cases iso_code wise as a horizontal bar chart (adjust the figure size so that the plot is clean and shows all the countries).
 - (b) Plot the total deaths iso_code wise as a horizontal bar chart (adjust the figure size so that the plot is clean and shows all the countries).

Submission: Put your codes in a python notebook and name it Assignment5. Before submitting your work to eLC, restart the kernel, clear all outputs and run it from top to bottom to ensure that you are getting the desired results. As always, print out the results of your code with a helpful message.